Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 6 - Alignment Recorrection, Dimension Matching, Basic Filtering.

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I. Work done this Week:-

This week we basically worked on the dimension alignment correction, like mentioned in the previous report, we were facing major issues with the dimensions between multiple scans, causing our subtraction method not to work properly due to the dimension mis match while going through each scan, The Mathematics regarding the dimension alignment has been mentioned in one of the latter paragraphs.

After we were able to match the multiple dimensions of these scans, we noticed that the volume was not matching but the machine had began working as same dimension of the voxels could perform the subtraction operation compared to before, but accuracy was not that high, and compared to our previous results, the skull from the scan was also being included in each subtraction analysis due to the upscaling, it was causing false positive readings due to the different bone densities throughout the skull, but many other portions of the brain matter was also being labelled as tumours whereas many actual regions were being missed out on by the current method.

So Accordingly we are planning on changing our methodology, We have currently tried using the Laplacian Filter, but that has not been able to remove the false positives and false negatives, messing up any form of accuracy within our system, but we can feel that it is working because the difference is able to detect many regions which do in fact contain the tumor, we have specifically ran this on one of our test scans RS005, attached below for reference, which shows that it is detecting multiple regions with tumor, but also flagging the skulls and eye sockets as readings, which has to be corrected and its missing out on some of the tumor region.

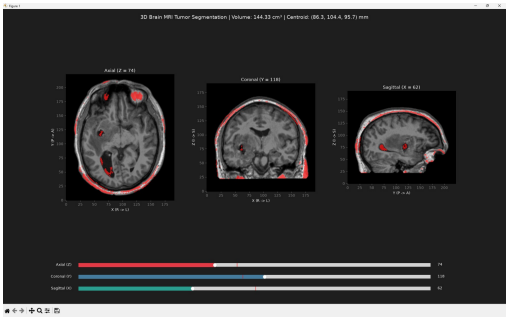


Fig. 1. Result

A. Brain Isolation and Preprocessing:

On T1-weighted MRI scans, non-brain parts like scalp fat, skull bone, and neck muscles appear very bright. If we leave them in, the algorithm will confuse them with tumors during subtraction. We must isolate only the brain tissue. The following methods were used:

1) **Affine Alignment & 3D Resampling:**
Brains come in different orientations and non-cubic voxel shapes. We map the raw voxel index grid

$$x = [x, y, z, 1]^T$$

to real-world coordinates

$$X = [X, Y, Z, 1]^T$$

in millimeters using an affine matrix M:

$$X = Mx = \begin{bmatrix} A & T \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}$$

To make the voxels cubic without introducing false edges or blur, we use 3D trilinear interpolation between the 8 corner points of each voxel cube:

$$I(x,y,z) = \sum_{i=0}^{1}\sum_{j=0}^{1}\sum_{k=0}^{1} f(x_i,y_j,z_k)(1-|\Delta x-i|)(1-|\Delta y-j|)(1-|\Delta z-k|)$$

This creates uniform, cubic voxels so that our 3D filters work equally in all directions.

2) **Morphology & Distance Transform (Skull Stripping):** To cut thin bridges connecting the brain to the skull (like eye nerves and dural folds), we apply 3D erosion $A \ominus B$:

$$A \ominus B = \{x \in \mathbb{Z}^3 \mid B_x \subseteq A\}$$

This breaks narrow connections, splitting the volume into separate pieces. To find the true brain and ignore stray muscle, we locate the piece k^* that has the highest internal distance from the edges while staying near the center C_0:

$$k^* = \arg\max_k \left(\frac{\sum_{p \in C_k} \mathrm{EDT}(p)}{1 + \frac{\|c_k - c_0\|_2^2}{\sigma_{\mathrm{pos}}^2}} \right)$$

Here, $\mathrm{EDT}(p)$ is the Euclidean Distance Transform, which measures how deep each point is inside the tissue. We then expand the brain back to its natural boundary using dilation $A \oplus B$ and fill internal dark holes to protect the brain cortex.

3) **Centering and Intensity Scaling:** We calculate the center of mass of the brain mask $\mathcal{M}$:

$$\bar{x} = \frac{1}{\sum \mathcal{M}} \sum x\mathcal{M}$$

We slide the brain into the middle of our standard matrix using a translation shift

$$t = \left\lfloor \frac{D - C}{2} \right\rfloor,$$

then scale the intensities to $[0, I_{\max}]$ so every image shares the same brightness range.

B. Subtraction pipeline:

Our pipelines uses two levels of difference.

A. absolute difference between baseline images and difference between baseline with test image

B. absolute difference of the results produced in level 1

Directly subtracting the brain MRI scan of different people can lead to mismatched normal brain folds (the curves observed) light up as bright spots, leading to false positives. We use multi-step subtraction and symmetry checks to isolate true tumors from normal brain differences.

1) **Combinatorial Baseline Variance:** We load N healthy brain scans into GPU memory and subtract every pair (i, j):

$$\Delta_{i,j}(x) = |B_i(x) - B_j(x)|$$

This gives

$$P = \frac{N(N-1)}{2}$$

unique difference maps. They tell us the normal variation

$$\tilde{\Delta}_{\mathrm{base}}(x)$$

expected in healthy people.

2) **3D Laplacian Edge Detection:** To detect sharp edges and set dynamic boundaries, we calculate second-order derivatives in 3D across the 6 direct neighbor voxels $\mathcal{N}_6$:

$$\nabla^2 f(x,y,z) = \frac{1}{h^2}\left(\sum_{p \in \mathcal{N}_6} f(p) - 6f(x,y,z) \right)$$

The average response sets a dynamic edge threshold

$$T_{\mathrm{dynamic}} = k \cdot \mu_L,$$

helping us separate true tumor boundaries from smooth background tissue.

3) **Two-Level Subtraction Gating:** We compare the test brain against the baseline in two passes:

   a) Level-1: Raw absolute difference:

$$D_1(x) = |V_{\mathrm{test}}(x) - B(x)|$$

   b) Level-2: Excess difference beyond normal healthy variation:

$$D_2(x) = \left| |V_{\mathrm{test}}(x) - B(x)| - \tilde{\Delta}_{\mathrm{base}}(x) \right|$$

   A voxel is only marked as a possible tumor if it is larger than the local threshold in both D_1 and D_2. This avoids false alarms in regions where healthy brains normally vary.

4) **Left-Right Symmetry Check:** Healthy human brains are mostly mirror-symmetric across the center line (mid-sagittal plane), but tumors almost always grow on only one side. We flip the brain horizontally using a reflection operator $\mathcal{R}_{\mathrm{MSP}}$ and subtract it from itself:

$$\Delta_{\mathrm{sym}}(x) = |V_{\mathrm{test}}(x) - V_{\mathrm{test}}(\mathcal{R}_{\mathrm{MSP}}(x))|$$

   Normal brain folds cancel out ($\Delta_{\mathrm{sym}} \approx 0$), while the tumor stands out clearly ($\Delta_{\mathrm{sym}} \gg 0$).

Code implementation for improving the segmentation and analysis of tumors.

II. Work to be done Next Week:-

Like Mentioned, the current filtering method is not working correctly, so we are planning on trying some of the other various filters that we have learnt in class, like the Gaussian and the other Arithmetic Filters to try to improve the accuracy, and remove the cases of false positives and false negatives, because the current reading has a bigger issue with false

positives, which a medical professional would prefer not to work with, so we have to further correct that more.

We also plan to change the window size accordingly because maybe that is also causing the errors through the multiple scans, we are currently using w of 3x3, we plan to check for the same with a window of size 5x5 and 7x7 (if necessary) or else, we are also going to compare with the other filters priorly mentioned, and we will proceed with the best possible method soon after, and try to calculate mathematically why the said method applies best, with these test cases we are trying to understand the rest of progression, this new revelation has been a boon for us, and we believe with correct filtering we are very close to the end of our project.

After we do this, we are planning on the GUI model once like mentioned last week, once basically complete the current model, we plan to look into various different methods to represent it into a more user friendly method so any person, who actually wants to use this project in a more productive way can directly do analysis via the interactive board, instead of dealing with the code, but we are also going to be completely transparent about our accuracy, this is our end of the project ambition, hoping that we can complete the rest of it before the deadline, because this as a end product output would be extremely helpful for society.

Repo Link:-
https://github.com/Rushin-Mod/DIP_2026_P20_G9↗